Week-8 | Summary

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1. Common

1.1. Notation

Scalars:

$$x_1, x_2, y_1, y_2, z_2, z_2, a, b, \alpha, \beta$$

Column vector:

$$\mathbf{x} \in \mathbb{R}^d$$

$$\mathbf{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_d \end{bmatrix}$$

Row vector:

$$\mathbf{x} \in \mathbb{R}^d$$

$$\mathbf{x}^T = \begin{bmatrix} x_1 & \cdots & x_d \end{bmatrix}$$

Matrix:

$$\mathbf{X} \in \mathbb{R}^{d \times n}$$

1.2. Labeled Dataset

$$D = \{(\mathbf{x}_1, y_1), \ \cdots, (\mathbf{x}_n, y_n)\}$$

1.3. Data-matrix

$$\mathbf{X} \in \mathbb{R}^{d \times n}$$

- $d \rightarrow$ number of features
- $n \rightarrow$ number of data-points

$$\mathbf{X} = \begin{bmatrix} | & & | \\ \mathbf{x}_1 & \cdots & \mathbf{x}_n \\ | & & | \end{bmatrix}$$

1.4. Label Vector

$$\mathbf{y} \in \{0, 1\}^n$$

or

$$\mathbf{y} \in \{-1, 1\}^n$$

where

$$\mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$$

1.5. Data-point

$$\mathbf{x}_i \in \mathbb{R}^d$$

1.6. Label set

For binary classification, the label set could be $\{0, 1\}$ or $\{1, -1\}$.

2. Discriminative and Generative modeling

<u>Generative model</u>

Learn the joint distribution between features and labels:

$$P(\mathbf{x}, y) = P(y)P(\mathbf{x} \mid y)$$

Data generation can be understood as a two step process:

- Choose a class: this is governed by the prior $P(y)$
- Sample a data-point from the class-conditional distribution $P(\mathbf{x} \mid y)$

Generative models are powerful if the distribution chosen closely matches the true process generating the data.

<u>Discriminative model</u>

Learn the conditional distribution of the label given the data-point:

$$P(y \mid \mathbf{x})$$

Discriminative models do not worry about how the data-point is generated. It is sufficient to "discriminate" between the two classes.

3. Generative models

<u>Binary classification</u>

Consider a binary classification problem with binary features:

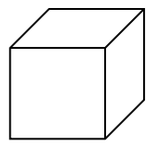
$$\mathbf{x}_i \in \{0,1\}^d, \quad y_i \in \{0,1\}$$

If the class-conditional independence is not assumed, the number of free parameters is given by:

$$2 \times \left(2^d - 1\right) + 1$$

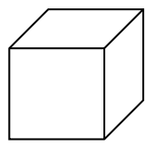
Visually:

Class-1



2^d faces

Class-2

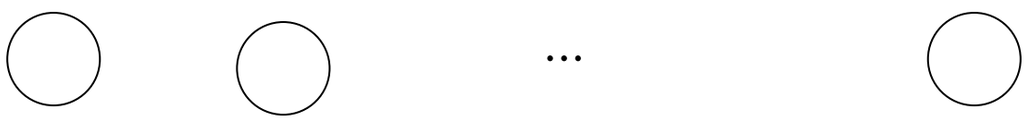


If the class-conditional independence is assumed, the number of free parameters is given by:

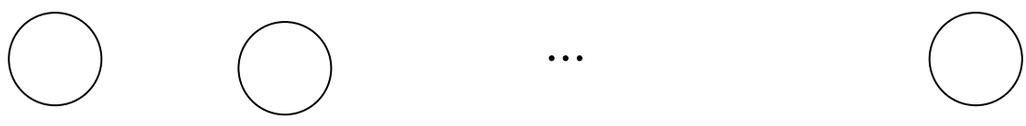
$$2 \times d + 1$$

Visually, d coins, one parameter per coin:

Class-1



Class-2



<u>Multi-class classification, discrete feature set</u>

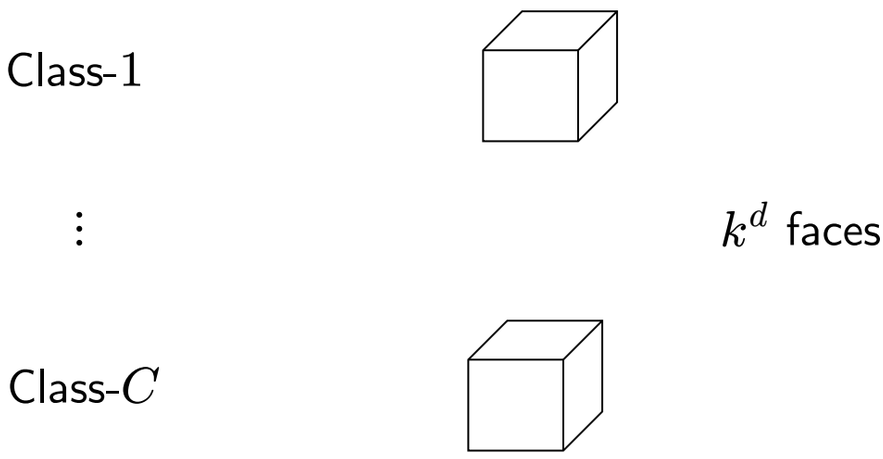
In general, if we have C classes:

$$\mathbf{x}_i \in \{0, \cdots, k-1\}^d, \; y_i \in \{1, \cdots, C\}$$

If the class-conditional independence is not assumed, the number of free parameters is given by:

$$C \times \left(k^d - 1\right) + (C - 1)$$

Visually:

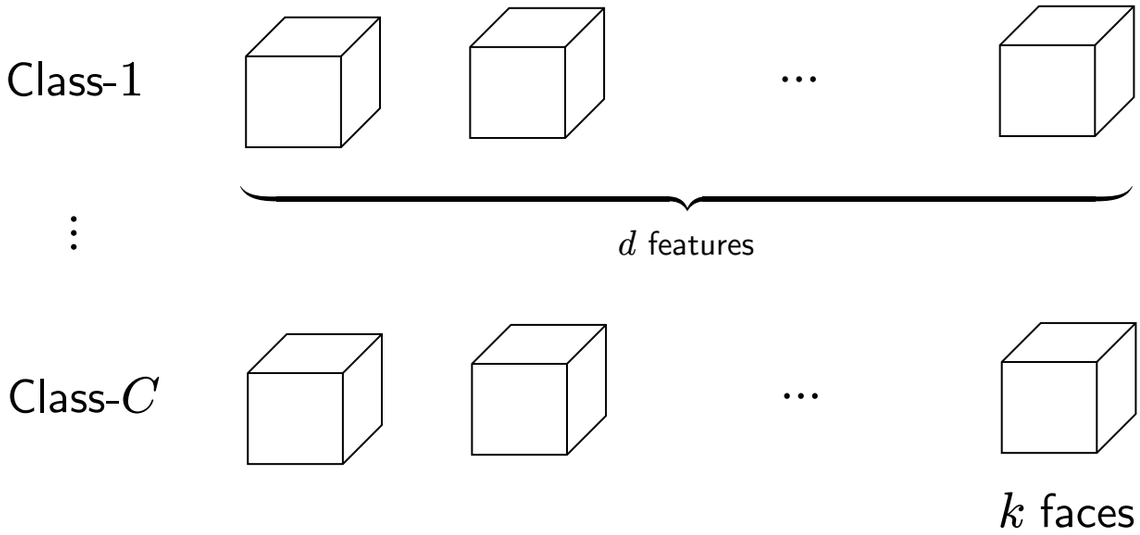


The number of parameters is exponential in the number of features.

If the class-conditional independence is assumed, the number of free parameters is given by:

$$C \times d(k-1) + (C - 1)$$

Visually:



4. Naive Bayes

<u>Notation</u>

Binary features, binary classification:

$$\mathbf{x}_i \in \{0,1\}^d, \; y_i \in \{0,1\}$$

$$x_{ij} = \begin{cases} 1, & \text{feature } j \text{ is present in sample } i \\ 0, & \text{feature } j \text{ is not present in sample } i \end{cases}$$

Parameter for the prior probability of the label:

$$p = P(Y = 1)$$

Parameter for feature j under class y:

$$p_j^y = P(X_j = 1 \mid Y = y)$$

Class-0	p_1^0	p_2^0	$\cdots$	p_d^0
Class-1	p_1^1	p_2^1	$\cdots$	p_d^1

<u>MLE</u>

For one-data point $(\mathbf{x}_i, y_i)$:

$$p^{y_i}(1-p)^{1-y_i}\prod_{j=1}^{d}\left(p_j^{y_i}\right)^{x_{ij}}\left(1-p_j^{y_i}\right)^{1-x_{ij}}$$

For the entire dataset:

$$L(\boldsymbol{\theta}; D) = \prod_{i=1}^{n}\left[p^{y_i}(1-p)^{1-y_i}\prod_{j=1}^{d}\left(p_j^{y_i}\right)^{x_{ij}}\left(1-p_j^{y_i}\right)^{1-x_{ij}}\right]$$

where $\boldsymbol{\theta}$ is the collection of all parameters.

<u>Log-Likelihood</u>

$$l(\boldsymbol{\theta}; D) = \sum_{i=1}^{n}\Big[y_i \log p + (1-y_i)\log(1-p)^{1-y_i} + \sum_{j=1}^{d} x_{ij}\log p_j^{y_i} + (1-x_{ij})\log\left(1-p_j^{y_i}\right)\Big]$$

<u>Estimates</u>

$$p = \frac{\sum_{i=1}^{n} \mathbb{I}[y_i = 1]}{n}$$

$$p_j^y = \frac{\sum_{i=1}^{n} \mathbb{I}[y_i = y] x_{ij}}{\sum_{i=1}^{n} \mathbb{I}[y_i = y]}$$

<u>Smoothing</u>

- $p_j^y = 0$ when no data-point in class y has $x_{ij} = 1$. Alternatively, every data-point in class y has $x_{ij} = 0$. This will create problems during prediction. To fix this, you add a data-point where all the features are 1 to both the classes.
- $p_j^y = 1$ when no data-point in class y has $x_{ij} = 0$. Alternatively, every data-point in class y has $x_{ij} = 1$. This will create problems during prediction. To fix this, you add a data-point where all the features are 0 to both the classes.

You end up adding four data-points to the entire dataset, two in each class. This method is called Laplace smoothing. The numerator increases by 1 and the denominator increases by 2.

<u>Prediction</u>

Using Bayes' rule:

$$P(y = 1 \mid \mathbf{x}_{\text{test}}) = \frac{P(y = 1) \cdot P(\mathbf{x}_{\text{test}} \mid y = 1)}{P(\mathbf{x}_{\text{test}})}$$

$$P(y=0 \mid \mathbf{x}_{\text{test}}) = \frac{P(y=0) \cdot P(\mathbf{x}_{\text{test}} \mid y=0)}{P(\mathbf{x}_{\text{test}})}$$

If we only need to find the predicted label and not the actual probability, we just need to compute the numerators and compare their magnitudes:

$$\widehat{y} = \begin{cases} 1, & P(y=1) \cdot P(\mathbf{x}_{\text{test}} \mid y=1) \geqslant P(y=0) \cdot P(\mathbf{x}_{\text{test}} \mid y=0) \\ 0, & \text{otherwise} \end{cases}$$

<u>Decision Boundary</u>

$$\frac{P(y=1 \mid \mathbf{x})}{P(y=0 \mid \mathbf{x})} = 1$$

The decision boundary is linear:

$$\mathbf{w}^T \mathbf{x} + b = 0$$

where:

$$w_j = \log\left[\frac{p_j^1\left(1-p_j^0\right)}{p_j^0\left(1-p_j^1\right)}\right]$$

$$b = \log\left(\frac{p}{1-p}\right) + \sum_{j=1}^{d} \log\left(\frac{1-p_j^1}{1-p_j^0}\right)$$

5. Linear Discriminant Analysis (Gaussian Naive Bayes)

This is a generative model where all the features are continuous. We

assume that the data for a given class is drawn from a multi-variate Gaussian. The two classes are assumed to share the same covariance matrix.

<u>Class conditional density</u>

$$\mathbf{x} \mid y = 0 \sim \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma})$$

$$P(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^d |\boldsymbol{\Sigma}|}} \exp\left[-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_0)^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}_0)\right]$$

$$\mathbf{x} \mid y = 1 \sim \mathcal{N}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma})$$

$$P(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^d |\boldsymbol{\Sigma}|}} \exp\left[-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_1)^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}_1)\right]$$

<u>Estimates from MLE</u>

$$\boldsymbol{\mu}_1 = \frac{\sum_{i=1}^{n} \mathbb{I}[y_i = 1]\mathbf{x}_i}{\sum_{i=1}^{n} \mathbb{I}[y_i = 1]}, \quad \boldsymbol{\mu}_0 = \frac{\sum_{i=1}^{n} \mathbb{I}[y_i = 0]\mathbf{x}_i}{\sum_{i=1}^{n} \mathbb{I}[y_i = 0]}$$

$$\boldsymbol{\Sigma} = \frac{1}{n} \sum_{i=1}^{n} (\mathbf{x}_i - \boldsymbol{\mu}_{y_i})(\mathbf{x}_i - \boldsymbol{\mu}_{y_i})^T$$

<u>Decision Boundary</u>

$$\mathbf{w}^T\mathbf{x} + b = 0$$

$$\mathbf{w} = (\boldsymbol{\mu}_0 - \boldsymbol{\mu}_1)^T \boldsymbol{\Sigma}^{-1}$$

$$b = \frac{1}{2}\left[\boldsymbol{\mu}_1^T \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_1 - \boldsymbol{\mu}_0^T \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_0 + \log \frac{p}{1-p}\right]$$

The decision boundary is linear.